

第四节 多元复合函数的求导法则

一、多元复合函数的一阶偏导数的计算

二、多元复合函数的二阶偏导数的计算

三、全微分形式不变性

一、多元复合函数的一阶偏导数的计算

1. 一元函数与多元函数复合的情形

定理1 如果函数 $u = \varphi(t)$ 及 $v = \psi(t)$ 都在点 t 可导，函数 $z = f(u, v)$ 在对应点 (u, v) 具有连续偏导数，那么复合函数 $z = f(\varphi(t), \psi(t))$ 在点 t 可导，且有

$$\frac{dz}{dt} = \frac{\partial z}{\partial u} \cdot \frac{du}{dt} + \frac{\partial z}{\partial v} \cdot \frac{dv}{dt}.$$

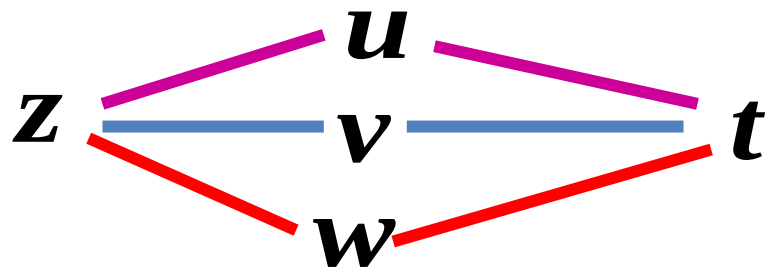
——全导数公式

一、多元复合函数的一阶偏导数的计算

定理的结论可推广到中间变量多于两个的情况.

例如, $z = f(u, v, w)$, $u = u(t)$, $v = v(t)$, $w = w(t)$,

则
$$\frac{dz}{dt} = \frac{\partial z}{\partial u} \cdot \frac{du}{dt} + \frac{\partial z}{\partial v} \cdot \frac{dv}{dt} + \frac{\partial z}{\partial w} \cdot \frac{dw}{dt}$$



例1 设 $z = 2uv + \sin w$, 其中 $u = \cos t, v = e^t, w = t^2$, 求 $\frac{dz}{dt}$.

解:

法一、
$$\begin{aligned}\frac{dz}{dt} &= \frac{\partial z}{\partial u} \cdot \frac{du}{dt} + \frac{\partial z}{\partial v} \cdot \frac{dv}{dt} + \frac{\partial z}{\partial w} \cdot \frac{dw}{dt} \\ &= 2v \cdot (-\sin t) + 2u \cdot e^t + \cos w \cdot 2t \\ &= 2e^t(\cos t - \sin t) + 2t \cos t^2.\end{aligned}$$

法二、 $\because z = 2e^t \cos t + \sin t^2$

$$\begin{aligned}\therefore \frac{dz}{dt} &= 2e^t \cdot (-\sin t) + 2e^t \cos t + 2t \cos t^2 \\ &= 2e^t(\cos t - \sin t) + 2t \cos t^2.\end{aligned}$$

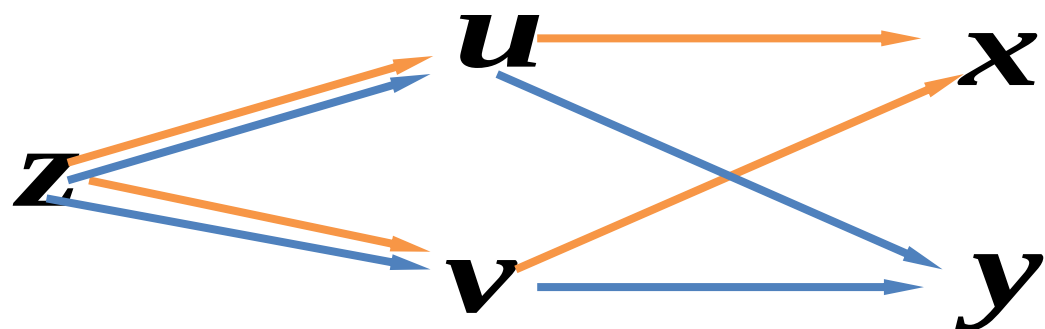
一、多元复合函数的一阶偏导数的计算

2. 多元函数与多元函数复合的情形

定理2 如果函数 $u = \varphi(x, y)$ 及 $v = \psi(x, y)$ 都在点 (x, y) 具有对 x 及对 y 的偏导数, $z = f(u, v)$ 在点 (u, v) 具有连续偏导数, 则复合函数 $z = f[\varphi(x, y), \psi(x, y)]$ 在点 (x, y) 处的两个偏导数都存在, 且有

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x}, \quad \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y}.$$

链式法则如图示



$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x},$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y}.$$

可写为: $z_x = f_u \cdot u_x + f_v \cdot v_x, \quad z_y = f_u \cdot u_y + f_v \cdot v_y.$

或: $z_x = f'_1 \cdot u_x + f'_2 \cdot v_x, \quad z_y = f'_1 \cdot u_y + f'_2 \cdot v_y.$

例 2 设 $z = (x^2 + y^2)^{\sin(x+3y)}$, 求 z_x 和 z_y .

解 令 $u = x^2 + y^2$, $v = \sin(x + 3y)$, 则 $z = u^v$,

$$z_x = z_u \cdot u_x + z_v \cdot v_x$$

$$= v u^{v-1} \cdot 2x + u^v \ln u \cdot \cos(x + 3y)$$

$$= (x^2 + y^2)^{\sin(x+3y)} \left[\frac{2x \sin(x + 3y)}{x^2 + y^2} + \ln(x^2 + y^2) \cos(x + 3y) \right]$$

$$z_y = z_u \cdot u_y + z_v \cdot v_y$$

$$= v u^{v-1} \cdot 2y + u^v \ln u \cdot 3 \cos(x + 3y)$$

$$= (x^2 + y^2)^{\sin(x+3y)} \left[\frac{2y \sin(x + 3y)}{x^2 + y^2} + 3 \ln(x^2 + y^2) \cos(x + 3y) \right]$$

例 3 设 $z = f(xy, \frac{x}{y})$, f 可微, 求 z_x 和 z_y .

解 设 $u = xy$, $v = \frac{x}{y}$, 则 $z = f(u, v)$,

$$z_x = f_u \cdot u_x + f_v \cdot v_x = f_u \cdot y + f_v \cdot \frac{1}{y}$$

$$z_y = f_u \cdot u_y + f_v \cdot v_y = f_u \cdot x - f_v \cdot \frac{x}{y^2}$$

例 3 设 $z = f(xy, \frac{x}{y})$, f 可微, 求 z_x 和 z_y .

解 $z_x = f'_1 \cdot y + f'_2 \cdot (\frac{1}{y}) = y f'_1 + \frac{1}{y} f'_2.$

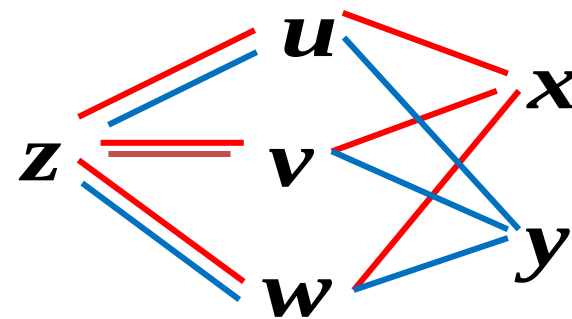
$$z_y = f'_1 \cdot x + f'_2 \cdot (-\frac{1}{y^2}) = x f'_1 - \frac{x}{y^2} f'_2.$$

定理的结论可推广到中间变量多于两个的情况.

(2) 设 $u = \varphi(x, y)$ 、 $v = \psi(x, y)$ 、 $w = w(x, y)$ 都在点 (x, y) 具有对 x 和 y 的偏导数, 复合函数 $z = f[\varphi(x, y), \psi(x, y), w(x, y)]$ 在对应点 (x, y) 的两个偏导数存在, 且可用下列公式计算:

$$z_x = f_u \cdot u_x + f_v \cdot v_x + f_w \cdot w_x ,$$

$$z_y = f_u \cdot u_y + f_v \cdot v_y + f_w \cdot w_y .$$



一、多元复合函数的一阶偏导数的计算

3. 其他情形

定理3 如果函数 $u = \varphi(x, y)$ 在点 (x, y) 具有对 x 及 y 的偏导数, 函数 $v = \psi(y)$ 在点 y 可导, 函数 $z = f(u, v)$ 在对应点 (u, v) 具有连续偏导数, 那么复合函数 $z = f(\varphi(x, y), \psi(y))$ 在点 (x, y) 处的两个偏导数都存在, 且有

$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x}, \quad \frac{\partial z}{\partial y} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial f}{\partial v} \cdot \frac{dv}{dy}.$$

特殊地, (1) $z = f(u, x, y)$ 其中 $u = \varphi(x, y)$

即 $z = f[\varphi(x, y), x, y]$,

$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial f}{\partial x},$$

$$\frac{\partial z}{\partial y} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial f}{\partial y}.$$

两者的区别

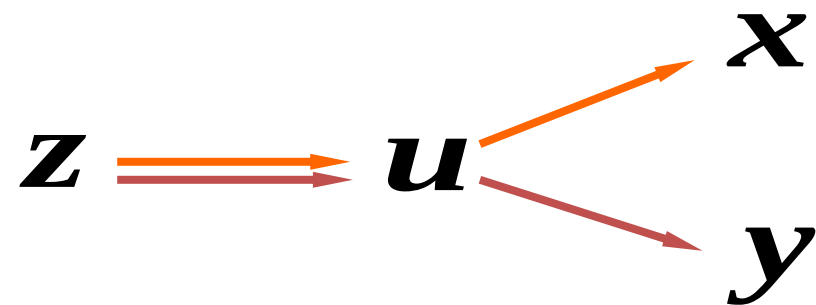
把复合函数 $z = f[\varphi(x, y), x, y]$ 中的 y 看作不变而对 x 的偏导数

把 $z = f(u, x, y)$ 中的 u 及 y 看作不变而对 x 的偏导数

(2) $z = f(u)$ 可微, 其中 $u = \varphi(x, y)$ 具有偏导数,

则 $z = f[\varphi(x, y)]$ 有偏导数, 且有:

$$\frac{\partial z}{\partial x} = \frac{df}{du} \cdot \frac{\partial u}{\partial x}, \quad \frac{\partial z}{\partial y} = \frac{df}{du} \cdot \frac{\partial u}{\partial y}.$$



例 4 设 $z = f(\frac{y}{x})$, f 可微, 证明 $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 0$.

证明 设 $u = \frac{y}{x}$, 则 $z = f(u)$,

$$\frac{\partial z}{\partial x} = \frac{df}{du} \cdot \frac{\partial u}{\partial x} = f'(u) \cdot \left(-\frac{y}{x^2}\right) = -\frac{y}{x^2} f'(u),$$

$$\frac{\partial z}{\partial y} = \frac{df}{du} \cdot \frac{\partial u}{\partial y} = f'(u) \cdot \left(\frac{1}{x}\right) = \frac{1}{x} f'(u),$$

$$\therefore x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 0.$$

例5 设 $z = f(x^2, e^{2x})$, f 可微, 求 $\frac{dz}{dx}$.

例6 设 $z = f(x, x^2 + y^2, xy)$, f 可微, 求 $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}$.

一、多元复合函数的一阶偏导数的计算

注记

- (1) 对于多元复合函数求偏导数，首先要借助结构图，弄清复合关系，哪些是自变量，哪些是中间变量.
- (2) 对于某个自变量求偏导数时，要经过一切与其有关的中间变量，最后归结到该自变量.

二、多元复合函数的二阶偏导数的计算

例 7 设 $w = f(x + y + z, xyz)$, f 具有二阶

连续偏导数, 求 $\frac{\partial w}{\partial x}$ 和 $\frac{\partial^2 w}{\partial x \partial z}$.

解: 令 $u = x + y + z$, $v = xyz$;

$$\text{记 } f'_1 = \frac{\partial f(u, v)}{\partial u}, \quad f''_{12} = \frac{\partial^2 f(u, v)}{\partial u \partial v},$$

同理有 f'_2 , f''_{11} , f''_{22} .

$$\frac{\partial w}{\partial x} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial x} = f'_1 + yz f'_2;$$

$$\frac{\partial^2 w}{\partial x \partial z} = \frac{\partial}{\partial z} (f_1' + yzf_2') = \frac{\partial f_1'}{\partial z} + yf_2' + yz \frac{\partial f_2'}{\partial z};$$

$$\frac{\partial f_1'}{\partial z} = \frac{\partial f_1'}{\partial u} \cdot \frac{\partial u}{\partial z} + \frac{\partial f_1'}{\partial v} \cdot \frac{\partial v}{\partial z} = f_{11}'' + xyf_{12}'';$$

$$\frac{\partial f_2'}{\partial z} = \frac{\partial f_2'}{\partial u} \cdot \frac{\partial u}{\partial z} + \frac{\partial f_2'}{\partial v} \cdot \frac{\partial v}{\partial z} = f_{21}'' + xyf_{22}'';$$

$$\begin{aligned} \text{于是 } \frac{\partial^2 w}{\partial x \partial z} &= f_{11}'' + xyf_{12}'' + yf_2' + yz(f_{21}'' + xyf_{22}'') \\ &= f_{11}'' + y(x+z)f_{12}'' + xy^2zf_{22}'' + yf_2'. \end{aligned}$$

注意: 对抽象函数 $f(u, v)$, 其偏导数 f_u 、 f_v 或 f_1' 、 f_2' 均仍然是多元复合函数, 它们与 f 具有相同的中间变量和相同的自变量.

例8 已知 $z = f(xy, x^2 - y^2)$, 其中 $f \in C^{(2)}$, 求 $\frac{\partial^2 z}{\partial x^2}$ 和 $\frac{\partial^2 z}{\partial x \partial y}$.

[解] 令 $u = xy, v = x^2 - y^2$, 则 $z = f(u, v)$

$$\frac{\partial^2 z}{\partial x^2} = y^2 f_{uu}(u, v) + 4xy f_{uv}(u, v) + 4x^2 f_{vv}(u, v) + 2 f_v(u, v).$$

$$\frac{\partial^2 z}{\partial x \partial y} = f_u + xy f_{uu} + 2(x^2 - y^2) f_{uv} - 4xy f_{vv}.$$

例9 已知 $u = xy + z f\left(\frac{\ln y}{x}\right)$, 其中 $f \in C^{(2)}$, 求 $\frac{\partial^2 u}{\partial x \partial y}$ 。

$$\frac{\partial^2 u}{\partial x \partial y} = 1 - \frac{z \ln y}{x^3 y} f''\left(\frac{\ln y}{x}\right) - \frac{z}{x^2 y} f'\left(\frac{\ln y}{x}\right)。$$

三、全微分形式不变性

设函数 $z = f(u, v)$ 具有连续偏导数，则有全微分 ；

$$\mathrm{d}z = \frac{\partial z}{\partial u} \mathrm{d}u + \frac{\partial z}{\partial v} \mathrm{d}v$$

当 $u = \varphi(x, y)$ 、 $v = \psi(x, y)$ 时，

$$\text{有 } \mathrm{d}z = \frac{\partial z}{\partial x} \mathrm{d}x + \frac{\partial z}{\partial y} \mathrm{d}y$$

$$\begin{aligned}
dz &= \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy \\
&= \left(\frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} \right) dx + \left(\frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial y} \right) dy \\
&= \frac{\partial z}{\partial u} \left(\frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \right) + \frac{\partial z}{\partial v} \left(\frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy \right) \\
&= \frac{\partial z}{\partial u} du + \frac{\partial z}{\partial v} dv.
\end{aligned}$$

全微分形式不变性的实质：

无论 z 是自变量 u 、 v 的函数或中间变量 u 、 v 的函数，它的全微分形式是一样的。

三、全微分形式不变性

例10 设函数 $z = f(x, y)$ 满足方程 $x - az = \varphi(y - bz)$, 其中

φ 为可微函数, a 和 b 为实常数, 证明: $a \frac{\partial z}{\partial x} + b \frac{\partial z}{\partial y} = 1$.

证明: 方程 $x - az = \varphi(y - bz)$ 两边求微分,

由全微分的不变性可知 $dx - a dz = \varphi'(y - bz)(dy - b dz)$,

$$\text{即 } dz = -\frac{1}{b\varphi'(y - bz) - a} dx + \frac{\varphi'(y - bz)}{b\varphi'(y - bz) - a} dy,$$

$$\text{故} \frac{\partial z}{\partial x} = -\frac{1}{b\varphi'(y-bz)-a}, \quad \frac{\partial z}{\partial y} = \frac{\varphi'(y-bz)}{b\varphi'(y-bz)-a},$$

$$\text{于是} a \frac{\partial z}{\partial x} + b \frac{\partial z}{\partial y} = \frac{-a}{b\varphi'(y-bz)-a} + \frac{b\varphi'(y-bz)}{b\varphi'(y-bz)-a} = 1.$$